



DBMS ASSIGNMENT 2

ER Diagram Design and Relational Schema

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Application: LibSphere — Library Management System

Objective

Design a complete Entity Relationship (ER) Diagram for the LibSphere Library Management System and convert it into a Relational Schema, including entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

1. Entities and Attributes

Categories

Attribute	Description	Key
category_id	Unique category ID	PK
category_name	Category/genre name	—

Authors

Attribute	Description	Key
author_id	Unique author ID	PK
first_name	Author first name	—
last_name	Author last name	—
bio	Author biography	—

Books

Attribute	Description	Key
book_id	Unique book ID	PK
title	Book title	—
isbn	ISBN number	—
category_id	Category assigned to book	FK
publication_year	Publication year	—
total_copies	Total copies	—
available_copies	Available copies	—

Book_Authors

Attribute	Description	Key
book_id	Book ID	PK, FK
author_id	Author ID	PK, FK

Members

Attribute	Description	Key
member_id	Unique member ID	PK
first_name	Member first name	—
last_name	Member last name	—
email	Member email	—
phone	Member phone	—
membership_date	Membership date	—
status	Membership status	—

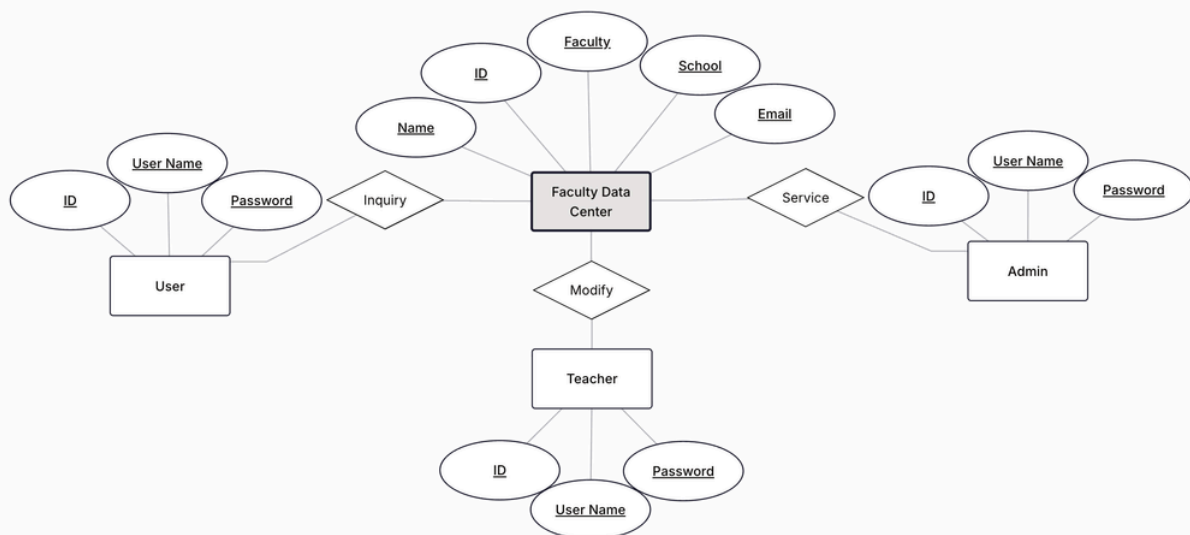
Loans

Attribute	Description	Key
loan_id	Unique loan ID	PK
book_id	Borrowed book ID	FK
member_id	Member ID	FK
loan_date	Loan/issue date	—
due_date	Due date	—
return_date	Return date	—
status	Loan status	—

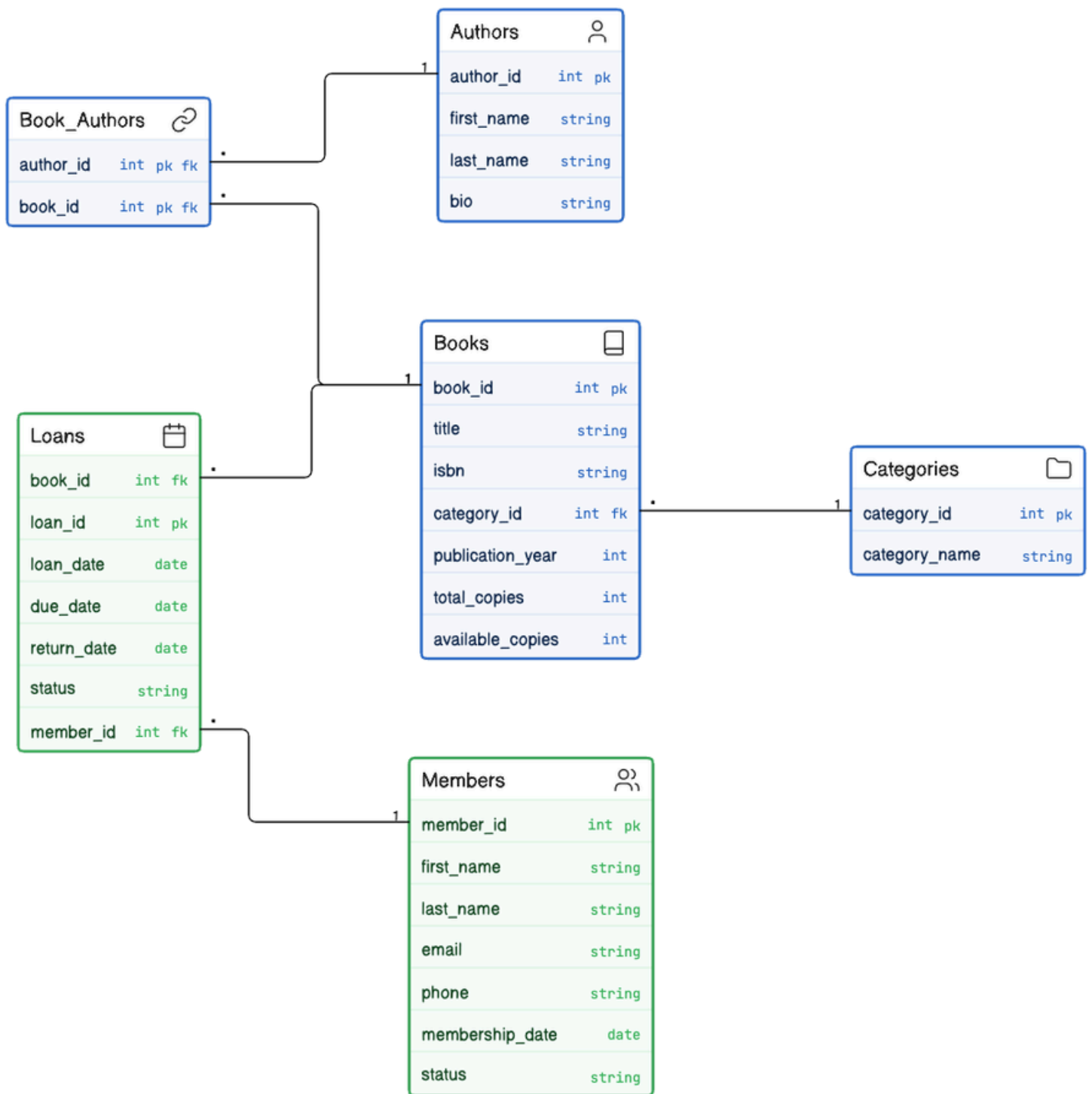
2. Relationships and Cardinality

Relationship	Cardinality	Explanation
Categories → Books	1 : M	One category can contain several books;each book is assigned to one category A
Books ↔ Authors	M : N	book may have multiple authors and an author may contribute to multiple book
Members → Loans	1 : M	One member can have multiple loan transactions; each loan belongs to one mem
Books → Loans	1 : M	A book title can be borrowed repeatedly over time; each loan refers to one book.

3. ER Diagram



4. Relational Schema



CATEGORY

```
CATEGORY(category_id PK, category_name)
```

AUTHOR

```
AUTHOR(author_id PK, first_name, last_name, bio)
```

BOOK

```
BOOK(book_id PK, title, isbn, category_id FK, publication_year, total_copies, available_copies)
```

BOOK_AUTHOR

```
BOOK_AUTHOR(book_idPK/FK, author_id PK/FK)
```

MEMBER

```
MEMBER(member_id PK, first_name, last_name, email, phone, membership_date, status)
```

LOAN

```
LOAN(loan_idPK,book_id FK, member_id FK, loan_date, due_date, return_date, status)
```

5. Primary and Foreign Keys

Table	Primary Key	Foreign Key
Categories	category_id	—
Authors	author_id	—
Books	book_id	category_id → Categories(category_id)
Book_Authors	book_id, author_id	book_id → Books(book_id); author_id → Authors(author_id)
Members	member_id	—
Loans	loan_id	book_id → Books(book_id); member_id → Members(member_id)

6. Relationship Explanation

Categories – Books (1:M)

A single category can be associated with several books, while each book is assigned to one category. The relationship is maintained through Books.category_id.

Books – Authors (M:N)

A book may have one or more authors, and an author may be associated with several books. The Book_Authors junction table implements this many-to-many relationship.

Members – Loans (1:M)

A member can borrow books multiple times during membership. Each loan transaction is associated with one member through Loans.member_id.

Books – Loans (1:M)

A book can be borrowed repeatedly by different members over time. Each individual loan transaction refers to one book through Loans.book_id.

7. Conclusion

TheLibSphere Library Management System uses six related entities: Categories, Authors, Books, Book_Authors, Members, and Loans. The design supports book cataloguing, author management, member registration, inventory tracking, and borrowing/return operations. Primary keys uniquely identify records, while foreign keys connect related entities. The Book_Authors junction table resolves the many-to-many relationship between Books and Authors.